

Rate, Leakage, and Distortion for Gaussian Sources with an Informed Encoder

Andrew H. Bond, *Senior Member, IEEE*

Abstract

An encoder observes a jointly Gaussian vector and describes part of it to a decoder that sees the description alone, under a weighted quadratic distortion. A second party holds a noisy linear observation of the source and never decodes. One description then carries two costs: its rate, and its conditional leakage to the second party, which is also the least rate at which that party can re-encode it. The rate–leakage region is known in single-letter form from secure source coding. We determine its structure for an informed encoder, which observes more than it must describe, when the decoder has no side information, for every distortion weight and every observation map of the second party. Both costs are convex in the error covariance of the description, so the whole frontier is a convex program with unique minimizers, and when the decoder’s and the second party’s geometries commute it is explicit at every budget. When the optimal description is one-dimensional, which an explicit test certifies, the minimum leakage is a closed-form function of the largest generalized eigenvalue of a matrix pencil in which the distortion budget enters, and the eigenvector is the direction the description reads. That direction turns with the budget unless the decoder’s and the second party’s geometries share it, whereas a description built for a single party reads a fixed direction. The rate-optimal and leakage-optimal descriptions differ whenever the second party’s information does not map the decoder’s principal direction onto itself. For independent components the leakage-optimal split of the distortion is not reverse water-filling and can describe different components from the rate-optimal split, and for a scalar described variable with a scalar context the pencil gives an explicit closed form.

Index Terms

Equivocation, Gaussian sources, information leakage, rate–distortion theory, secure source coding, side information.

I. INTRODUCTION

One description of data is often held by more than one party. The decoder that reconstructs from it pays its rate. A second party that holds the description together with its own observation of the source learns from it the conditional mutual information between the source and the description given that observation. In a secrecy reading the second party is an eavesdropper and this quantity is the information leakage beyond its side information; minimizing it maximizes the eavesdropper’s equivocation about the source. In a storage reading the second party keeps the description compressed against what it already knows, and the same quantity is the least rate at which it can re-encode the description. Either way one description carries two costs, set by two different parties, and the encoder, which may observe more of the source than either party, decides how they trade off.

The single-letter region of jointly achievable rate, distortion, and equivocation is due to Villard and Piantanida [1, Theorem 3], and for vector Gaussian sources Ekrem and Ulukus [2] obtained the minimum leakage when the rate is unconstrained, as an optimization over covariance matrices. The joint rate–leakage region was left open: they report that they could not solve the weighted problem that traces it [2, p. 8], and a characterization through new extremal inequalities has since been reported by Xu and Chen [5]. Without decoder side information the region itself is elementary, because a single Gaussian exhaustion lemma bounds both costs at once (Proposition 2). What this paper contributes is the structure of that region for an informed encoder. The encoder observes a jointly Gaussian vector T , the decoder must reproduce a linear part $Y = FT$ under a weighted quadratic distortion, and the second party observes $S = HT + U$ for an arbitrary matrix H . When H reaches the parts of T outside Y , the encoder holds information about the second party’s observation that the decoder does not need, and this is what makes the two costs disagree.

The results are structural as well as explicit. The coding theorem (Theorem 3) identifies the operational region with the single-letter one, and Gaussian descriptions, indexed by their error covariance, exhaust it (Proposition 2). In that coordinate both costs are convex (Theorem 4), so every point of the rate–leakage frontier is the unique minimizer of a convex program and the minimum leakage is a determinant maximization under linear matrix inequalities. When the optimal description is one-dimensional, which an explicit certificate decides, the minimum leakage is

$$L(D) = \frac{1}{2} \log(1 + \Delta/\mu_{\max}),$$

with Δ the reduction in distortion the description must buy and $\mu_{\max}$ the largest generalized eigenvalue of a pencil formed from the source covariance, the decoder’s weight, the conditional covariance left after the second party’s observation, and the budget (Theorem 8). For a scalar described variable and a scalar context this is an explicit closed form (Corollary 9). When the decoder’s and the holder’s geometries commute, the whole frontier is explicit at every budget (Theorem 6), and more generally the problem splits along every subspace decomposition that both geometries respect (Theorem 5).

The eigenvector of the pencil is the direction the description reads, and it carries the paper’s main structural finding. The rate-optimal description reads the principal direction of the decoder’s weighted covariance at every budget, as do principal components, reverse water-filling [3], and the Gaussian information bottleneck [4]. The leakage-optimal description reads a direction that turns with the budget, unless that direction is a common generalized eigenvector of the decoder’s geometry and the second party’s (Theorem 11). No description optimized for a single party does this (Proposition 10). The same condition governs whether the two optimal descriptions coincide at all (Theorem 13): in the one-dimensional regime they differ whenever the second party’s information does not map the decoder’s principal direction onto itself. For independent components the leakage-optimal split of the distortion equalizes slopes that we compute in closed form, activates components in an order that is not the order of their variances, and can describe a disjoint set of components from the rate-optimal split (Section VII).

Related work

The conditional leakage studied here is the secrecy literature’s quantity. For a deterministic encoder, $h(T^n | M, S^n) = h(T^n | S^n) - H(M | S^n)$, so minimizing $H(M | S^n)/n$, which equals $I(T^n; M | S^n)/n$, is the same as maximizing the equivocation $h(T^n | M, S^n)/n$. Secure lossy source coding with side information at the eavesdropper was characterized by Villard and Piantanida [1]; with no decoder side information their Theorem 3 gives the discrete region of Theorem 18 below. Ekrem and Ulukus [2] treat vector Gaussian sources with side information at both receivers under a matrix distortion on the whole source, give an outer bound that is tight without a rate constraint, and leave the joint region open. Our distortion constrains a weighted part of the source and leaves the rest unpenalized, the decoder has no side information, and the eavesdropper’s observation map is arbitrary. Xu and Chen [5] report a characterization of the rate–distortion–equivocation function of vector Gaussian secure source coding, and Günlü, Schaefer, Boche, and Poor [6] evaluate the lossy rate region for scalar Gaussian sources with a remote source and separate secrecy and privacy leakage. Neither treats an encoder that observes components it need not describe, a weighted partial distortion with an arbitrary observation map, or the structure results below. Related problems price the leakage of a description about side information [7], [8], [9] or the equivocation of the source in a cipher system [10], [11]. The leakage endpoint is Steinberg’s common-reconstruction functional [12] on the source as the encoder observes it, an augmentation noted for finite alphabets by Lapidoth, Malär, and Wigger [13]; a vector Gaussian common-reconstruction function with every component penalized was given by Lu and Xu [14]. Weighted quadratic distortion models a decoder that cares about some directions more than others, as in weighted rate–distortion [15], indirect source coding [16], and task-based quantization [17]. Parameterizing a Gaussian description by its error covariance is a standard device in vector Gaussian source coding [18], and the resulting endpoint program is a determinant maximization [19].

The paper is organized as follows. Section II sets up the model and the coding theorem, Section III proves the convex structure, Section V the one-dimensional regime, Section VI the results on directions and misalignment, and Section VII independent components. Section VIII gives instances and Section IX open problems. The appendices prove the coding theorem and a Gaussian exhaustion lemma and describe the numerical checks.

II. MODEL AND THE CODING THEOREM

Logarithms are base two unless written $\ln$. The encoder observes i.i.d. copies of $T \in \mathbb{R}^p$, jointly Gaussian and centered with $\Sigma_T \succ 0$. The described variable is $Y = FT \in \mathbb{R}^m$ for a matrix F of full row rank, and the decoder's distortion is $\mathbb{E}[(Y - \hat{Y})^\top W(Y - \hat{Y})]$ for a weight $W \succ 0$. Write

$$Q := F^\top W F \succeq 0, \quad D_{\max} := \text{tr}(Q\Sigma_T).$$

The second party, which we call the holder, observes $S = HT + U \in \mathbb{R}^r$ with $U \sim \mathcal{N}(0, \Sigma_U)$, $\Sigma_U \succ 0$, independent of T and of all coding randomness. Write

$$J := H^\top \Sigma_U^{-1} H \succeq 0, \quad \Sigma_{T|S} = (\Sigma_T^{-1} + J)^{-1},$$

the information that S carries about T and the conditional covariance it leaves.

A code consists of an encoder $f_n : \mathbb{R}^{p \times n} \rightarrow \mathcal{M}_n$, a finite set, and a decoder $g_n : \mathcal{M}_n \rightarrow \mathbb{R}^{m \times n}$. With $M = f_n(T^n)$ and $\hat{Y}^n = g_n(M)$, its rate, distortion, and leakage are

$$R_n = \frac{1}{n} \log |\mathcal{M}_n|, \quad D_n = \frac{1}{n} \sum_{i=1}^n \mathbb{E}[(Y_i - \hat{Y}_i)^\top W(Y_i - \hat{Y}_i)], \quad L_n = \frac{1}{n} H(M | S^n) = \frac{1}{n} I(T^n; M | S^n).$$

A pair (R, L) is achievable at distortion D when a sequence of codes has $\limsup R_n \leq R$, $\limsup L_n \leq L$, and $\limsup D_n \leq D$; the closure of the achievable set is $\mathcal{R}(D)$. Throughout $0 < D < D_{\max}$; at $D \geq D_{\max}$ the zero reproduction gives $R = L = 0$. Write $R_{\min}(D)$ and $L(D)$ for the least rate and the least leakage over $\mathcal{R}(D)$, and call a description rate-optimal or leakage-optimal when it attains one of them.

A test channel $p(\hat{y} | t)$ is admissible when its distortion is at most D ; the Markov chain $\hat{Y} - T - S$ holds because the description is formed without S . For a channel, let Σ_{e0} and Σ_e be the error covariances of the best linear estimates of T from $\hat{Y}$ and from $(\hat{Y}, S)$. Appendix B proves the following Gaussian exhaustion lemma.

Lemma 1. *Let $\hat{Y} \in \mathbb{R}^k$, for any k , be centered with finite second moments, let A be its regression matrix on T and $N' = \hat{Y} - AT$ its residual, and suppose $\text{Cov}(N', S) = 0$ and $\text{Cov } N' \succ 0$. Then $I(T; \hat{Y}) \geq \frac{1}{2} \log(\det \Sigma_T / \det \Sigma_{e0})$ and $I(T; \hat{Y} | S) \geq \frac{1}{2} \log(\det \Sigma_{T|S} / \det \Sigma_e)$, and the jointly Gaussian channel $AT + N$, $N \sim \mathcal{N}(0, \text{Cov } N')$ independent of (T, U) , has the same second moments and meets both bounds with equality.*

Proposition 2. *The union of the quadrants above $(I(T; \hat{Y}), I(T; \hat{Y} | S))$ over admissible channels equals the union, over $\Sigma_{e0} \in \mathcal{P}(D)$, of the quadrants above*

$$(r(\Sigma_{e0}), \ell(\Sigma_{e0})) = \left(\frac{1}{2} \log \frac{\det \Sigma_T}{\det \Sigma_{e0}}, \frac{1}{2} \log \frac{\det \Sigma_{T|S}}{\det \Sigma_e} \right), \quad \Sigma_e^{-1} = \Sigma_{e0}^{-1} + J, \quad (1)$$

where $\mathcal{P}(D) = \{\Sigma_{e0} : 0 \prec \Sigma_{e0} \preceq \Sigma_T, \text{tr}(Q\Sigma_{e0}) \leq D\}$.

Proof. Given an admissible channel, centering $\hat{Y}$ changes neither coordinate and does not raise the distortion. If $\text{Cov } N'$ is singular, either a noiseless linear functional of T is revealed and both coordinates are infinite, or a nonzero combination of the components of $\hat{Y}$ vanishes and may be dropped after an invertible linear map; so assume $\text{Cov } N' \succ 0$. The residual is uncorrelated with T by construction and with U by independence, hence with S , and Lemma 1 replaces the channel by $\hat{Y} = AT + N$ with the same distortion and no larger coordinates. For it, $\Sigma_{e0}^{-1} = \Sigma_T^{-1} + A^\top \Sigma_N^{-1} A$, so $0 \prec \Sigma_{e0} \preceq \Sigma_T$; the pair $(\hat{Y}, S)$ is a linear observation of T with independent noises, so $\Sigma_e^{-1} = \Sigma_{e0}^{-1} + J$; and the distortion is at least that of $\mathbb{E}[Y | \hat{Y}]$, which is $\text{tr}(Q\Sigma_{e0})$. Conversely, given $\Sigma_{e0} \in \mathcal{P}(D)$, factor $\Sigma_{e0}^{-1} - \Sigma_T^{-1} = G^\top G$, let $Z = GT + N$ with $N \sim \mathcal{N}(0, I)$ independent of (T, U) , and reproduce $\hat{Y} = \mathbb{E}[Y | Z]$. Then $\text{Cov}(T | Z) = \Sigma_{e0}$, the distortion is $\text{tr}(Q\Sigma_{e0}) \leq D$, $\hat{Y} - T - S$ holds, and since $\hat{Y}$ is a function of Z , data processing bounds its coordinates by those of Z , which are (1). $\square$

Theorem 3. *For every $0 < D < D_{\max}$, $\mathcal{R}(D)$ equals the union of Proposition 2.*

The proof is in Appendix A. Its discrete core is the region of Villard and Piantanida without decoder side information (Theorem 18); the Gaussian extension quantizes the side information, the source, and the reproduction and passes to the limit.

III. CONVEX STRUCTURE

Theorem 4. On $\{0 \prec \Sigma_{e0} \preceq \Sigma_T\}$, the rate r and the leakage ℓ of (1) are strictly convex functions of Σ_{e0} , and the distortion $\text{tr}(Q\Sigma_{e0})$ is linear. Consequently:

- (a) for every $\alpha \in [0, 1]$ the program $\min\{\alpha r(\Sigma_{e0}) + (1 - \alpha)\ell(\Sigma_{e0}) : \Sigma_{e0} \in \mathcal{P}(D)\}$ has a unique minimizer, the distortion constraint is active there, and the minima over α trace the Pareto frontier of $\mathcal{R}(D)$;
- (b) in the coordinate Σ_e the minimum leakage is a determinant maximization,

$$L(D) = \frac{1}{2} \log \det \Sigma_{T|S} - \frac{1}{2} \max\{\log \det \Sigma_e : 0 \prec \Sigma_e \preceq \Sigma_{T|S}, \text{tr}(Qh(\Sigma_e)) \leq D\}, \quad h(\Sigma_e) := (\Sigma_e^{-1} - J)^{-1}, \quad (2)$$

in which h is matrix-convex and the constraint is equivalent to a linear matrix inequality: with $J = B^\top B$, $\text{tr}(Qh(\Sigma_e)) \leq D$ if and only if some symmetric Z has $\text{tr}(QZ) \leq D$ and

$$\begin{bmatrix} Z - \Sigma_e & \Sigma_e B^\top \\ B \Sigma_e & I - B \Sigma_e B^\top \end{bmatrix} \succeq 0; \quad (3)$$

- (c) a feasible Σ_e is the maximizer in (2) if and only if $\text{tr}(Qh(\Sigma_e)) = D$ and there are $\varpi \geq 0$ and $\Theta \succeq 0$ with

$$\frac{1}{2} \Sigma_e - \varpi \Sigma_{e0} Q \Sigma_{e0} = \Sigma_e \Theta \Sigma_e, \quad \Theta(\Sigma_{T|S} - \Sigma_e) = 0, \quad \Sigma_{e0} = h(\Sigma_e), \quad (4)$$

and then $\varpi > 0$.

Proof. Convexity. $r = \frac{1}{2} \log \det \Sigma_T - \frac{1}{2} \log \det \Sigma_{e0}$ is strictly convex. For the leakage, write $J = B^\top B$ and note that $\Sigma_e = (\Sigma_{e0}^{-1} + B^\top B)^{-1}$ is the error covariance of the best linear estimate of a vector with covariance Σ_{e0} from its noisy image under B with unit noise. For every matrix L , the estimator with gain L has error covariance $E_L(\Sigma_{e0}) = (I - LB)\Sigma_{e0}(I - LB)^\top + LL^\top \succeq \Sigma_e$, with equality at the optimal gain, and E_L is affine in Σ_{e0} . A pointwise minimum of affine maps in the Löwner order, attained at each point, is matrix-concave: at $\Sigma_t = (1 - t)\Sigma_0 + t\Sigma_1$ with optimal gain L_t , $\Sigma_e(\Sigma_t) = E_{L_t}(\Sigma_t) = (1 - t)E_{L_t}(\Sigma_0) + tE_{L_t}(\Sigma_1) \succeq (1 - t)\Sigma_e(\Sigma_0) + t\Sigma_e(\Sigma_1)$. The map $\Sigma \mapsto -\log \det \Sigma$ is strictly convex and decreasing in the Löwner order, so $-\log \det \Sigma_e(\Sigma_t) \leq -\log \det((1 - t)\Sigma_e(\Sigma_0) + t\Sigma_e(\Sigma_1)) \leq (1 - t)(-\log \det \Sigma_e(\Sigma_0)) + t(-\log \det \Sigma_e(\Sigma_1))$, with strict inequality in the second step when $\Sigma_0 \neq \Sigma_1$, because $\Sigma_{e0} \mapsto \Sigma_e$ is injective. So ℓ is strictly convex.

(a) $\mathcal{P}(D)$ is convex, the objective is strictly convex and diverges as $\det \Sigma_{e0} \rightarrow 0$, and the feasible set is bounded, so the minimizer exists and is unique. Both r and ℓ decrease when Σ_{e0} increases in the Löwner order, so at a minimizer with slack distortion a move toward Σ_T would stay feasible and lower the objective, since Σ_T itself has distortion $D_{\max} > D$. The region is convex and upward closed, so its Pareto points are exactly the minima of the weighted programs.

(b) The map $\Sigma_{e0} \mapsto \Sigma_e$ is a bijection of $\{0 \prec \Sigma_{e0} \preceq \Sigma_T\}$ onto $\{0 \prec \Sigma_e \preceq \Sigma_{T|S}\}$, with inverse $h: \Sigma_{e0} \preceq \Sigma_T$ if and only if $\Sigma_e^{-1} = \Sigma_{e0}^{-1} + J \succeq \Sigma_T^{-1} + J = \Sigma_{T|S}^{-1}$. The matrix $B\Sigma_{T|S}B^\top$ has the same nonzero eigenvalues as $\Sigma_{T|S}^{1/2} J \Sigma_{T|S}^{1/2} \prec \Sigma_{T|S}^{1/2}(\Sigma_T^{-1} + J)\Sigma_{T|S}^{1/2} = I$, so on the feasible set $B\Sigma_e B^\top \prec I$, and the Woodbury identity gives $h(\Sigma_e) = \Sigma_e + \Sigma_e B^\top (I - B\Sigma_e B^\top)^{-1} B \Sigma_e$. By the Schur complement criterion [20, Sec. A.5.5], $h(\Sigma_e) \preceq Z$ if and only if (3) holds, and the block matrix there is affine in (Σ_e, Z) ; averaging two instances with $Z_i = h(\Sigma_e^{(i)})$ gives matrix convexity of h . Since $\text{tr}(QZ)$ is monotone in Z , the constraint $\text{tr}(Qh(\Sigma_e)) \leq D$ is equivalent to the existence of Z with $\text{tr}(QZ) \leq D$ and (3).

(c) The program (2) has a strictly feasible point ($\Sigma_{e0} = \epsilon \Sigma_T$ with $\epsilon D_{\max} < D$), so the Karush–Kuhn–Tucker conditions are necessary and sufficient. The differential of h is $dh = h \Sigma_e^{-1} d\Sigma_e \Sigma_e^{-1} h$, so the gradient of $\text{tr}(Qh)$ is $\Sigma_e^{-1} \Sigma_{e0} Q \Sigma_{e0} \Sigma_e^{-1}$, and stationarity of $-\frac{1}{2} \ln \det \Sigma_e + \varpi \text{tr}(Qh) + \text{tr}(\Theta(\Sigma_e - \Sigma_{T|S}))$, multiplied on both sides by Σ_e , is the first condition of (4), the factor $\ln 2$ being absorbed in ϖ . When the distortion constraint is active its complementary slackness holds for every $\varpi \geq 0$, and the maximizer has an active constraint by (a). If $\varpi = 0$, then $\Theta = \frac{1}{2} \Sigma_e^{-1} \succ 0$ and the second condition forces $\Sigma_e = \Sigma_{T|S}$, which is infeasible; so $\varpi > 0$. $\square$

IV. DECOUPLING AND COMMUTING GEOMETRY

Whiten by the source covariance: write $\tilde{Q} = \Sigma_T^{1/2} Q \Sigma_T^{1/2}$, $\tilde{J} = \Sigma_T^{1/2} J \Sigma_T^{1/2}$, and $X = \Sigma_T^{-1/2} \Sigma_{e0} \Sigma_T^{-1/2}$. Since $\Sigma_{T|S} = \Sigma_T^{1/2} (I + \tilde{J})^{-1} \Sigma_T^{1/2}$ and $\Sigma_{e0}^{-1} + J = \Sigma_T^{-1/2} (X^{-1} + \tilde{J}) \Sigma_T^{-1/2}$, the costs and the constraint of Proposition 2 become

$$r = -\frac{1}{2} \log \det X, \quad \ell = \frac{1}{2} \log \det (X^{-1} + \tilde{J}) - \frac{1}{2} \log \det (I + \tilde{J}), \quad \text{tr}(\tilde{Q}X) \leq D, \quad 0 \prec X \preceq I. \quad (5)$$

Theorem 5. Suppose $\mathbb{R}^p = \bigoplus_k \mathcal{V}_k$ is an orthogonal decomposition into subspaces invariant under both $\tilde{Q}$ and $\tilde{J}$. Then for every $\alpha \in [0, 1]$ the minimizer of Theorem 4(a), written in the coordinate X , is block-diagonal with respect to the decomposition, and the weighted program splits into one program per block, coupled only through an allocation $D = \sum_k D_k$ that equalizes the slopes of the blocks' optimal values.

Proof. Let P_k be the orthogonal projector onto $\mathcal{V}_k$, and for signs $\epsilon_k \in \{\pm 1\}$ let $S_\epsilon = \sum_k \epsilon_k P_k$, an orthogonal symmetric matrix that commutes with $\tilde{Q}$ and $\tilde{J}$. The map $X \mapsto S_\epsilon X S_\epsilon$ preserves $\log \det X$, $\log \det(X^{-1} + \tilde{J})$ because $(S_\epsilon X S_\epsilon)^{-1} + \tilde{J} = S_\epsilon(X^{-1} + \tilde{J})S_\epsilon$, the distortion $\text{tr}(\tilde{Q} S_\epsilon X S_\epsilon) = \text{tr}(\tilde{Q} X)$, and the constraint $X \preceq I$. The minimizer is unique, so it is fixed by every S_ϵ , and then $P_k X P_l = \epsilon_k \epsilon_l P_k X P_l$ for all signs, which forces $P_k X P_l = 0$ for $k \neq l$. For block-diagonal X both costs and the distortion are sums over blocks, and the remaining problem is a separable convex program with one linear constraint, whose optimality condition is a common subgradient level. $\square$

The independent components of Section VII are the case of two-dimensional blocks, each holding one described coordinate and its context. When the blocks are one-dimensional, which happens exactly when $\tilde{Q}$ and $\tilde{J}$ commute, the whole region is explicit.

Theorem 6. Suppose $Q \Sigma_T J = J \Sigma_T Q$, equivalently $\tilde{Q} \tilde{J} = \tilde{J} \tilde{Q}$, and let $u_1, \dots, u_p$ be a common orthonormal eigenbasis with $\tilde{Q} u_i = q_i u_i$ and $\tilde{J} u_i = j_i u_i$. For $\alpha \in [0, 1]$ the frontier point of weight α is $X^* = \sum_i x_i u_i u_i^\top$, where $x_i = 1$ if $q_i = 0$ and otherwise $x_i = \min\{1, \xi_i(\theta)\}$, with $\xi_i(\theta)$ the positive root of

$$\theta q_i j_i x^2 + (\theta q_i - \alpha j_i) x - 1 = 0 \quad (6)$$

($\xi_i = 1/(\theta q_i)$ when $j_i = 0$), and $\theta > 0$ the unique level with $\sum_i q_i x_i = D$. Its rate and leakage are

$$R = \sum_i \frac{1}{2} \log \frac{1}{x_i}, \quad L = \sum_i \frac{1}{2} \log \frac{1 + j_i x_i}{x_i(1 + j_i)}.$$

In particular the minimum leakage takes $x_i = \min\{1, (\sqrt{1 + 4j_i/(\theta q_i)} - 1)/(2j_i)\}$, and direction i is described exactly when $\theta > (1 + \alpha j_i)/(q_i(1 + j_i))$, so the leakage-optimal description begins with the directions of largest $q_i(1 + j_i)$, the rate-optimal one with those of largest q_i .

Proof. By Theorem 5 with the one-dimensional blocks spanned by the u_i , the minimizer is $\sum_i x_i u_i u_i^\top$ and the program is $\min \sum_i \phi_i(x_i)$ subject to $\sum_i q_i x_i \leq D$ and $0 < x_i \leq 1$, with $\phi_i(x) = \frac{\alpha}{2} \ln \frac{1}{x} + \frac{1-\alpha}{2} \ln \frac{1+j_i x}{x(1+j_i)}$ in natural units. Each ϕ_i is convex and decreasing, with $-\phi'_i(x) = (1 + \alpha j_i)/(2x(1 + j_i x))$. A direction with $q_i = 0$ does not enter the constraint, so $x_i = 1$. For the others the optimality condition is $-\phi'_i(x_i) = \frac{1}{2} \theta q_i$ when $x_i < 1$ and $-\phi'_i(1) \leq \frac{1}{2} \theta q_i$ when $x_i = 1$, for a common $\theta > 0$; the first is (6), whose roots have product $-1/(\theta q_i j_i) < 0$, so exactly one is positive, and the second reads $\theta \leq (1 + \alpha j_i)/(q_i(1 + j_i))$. Each $x_i(\theta)$ is continuous and nonincreasing, strictly decreasing where it is below one, so $\sum_i q_i x_i(\theta)$ decreases continuously and strictly from $\sum_i q_i > D$ and θ is unique. The formulas for R and L are (5) at a diagonal X . $\square$

In each direction the leakage $\frac{1}{2} \log[(1 + j_i x_i)/(x_i(1 + j_i))]$ is Steinberg's scalar Gaussian common-reconstruction function [12] at unit variance, distortion x_i , and squared correlation $j_i/(1 + j_i)$ between the direction and the holder's observation; the theorem is the allocation of the distortion budget among such problems, and the joint frontier that goes with it.

Corollary 7. In commuting geometry every optimal description, at every budget and weight, reads the fixed directions u_i , consistent with Theorem 11. The rate-optimal and leakage-optimal descriptions nonetheless differ whenever two directions described by the rate-optimal one have different ratios j_i/q_i .

Proof. The first statement is Theorem 6. If the two optimal descriptions coincided, they would describe the same directions with the same x_i ; for those directions $x_i = 1/(\theta_R q_i)$ and $x_i(1 + j_i x_i) = 1/(\theta_L q_i)$, so $1 + j_i x_i = \theta_R/\theta_L$ is the same for all of them, and $j_i x_i = j_i/(\theta_R q_i)$ forces j_i/q_i to be the same. $\square$

V. THE ONE-DIMENSIONAL REGIME

Write $\Delta = D_{\max} - D$ for the reduction in distortion the description must buy, and

$$A(\Delta) = \Sigma_T Q \Sigma_T - \Delta \Sigma_T.$$

A one-dimensional description is $Z = b^\top T + N$ with $b \neq 0$ and $N \sim \mathcal{N}(0, \sigma^2)$ independent, reproduced by $\mathbb{E}[Y | Z]$. Its read direction is b , and the direction in the space of the described variable along which it reduces the error is $F \Sigma_T b$.

Theorem 8. (a) A one-dimensional description meets the distortion constraint with finite leakage if and only if $\Delta < \lambda_{\max}(W^{1/2} \Sigma_Y W^{1/2})$, where $\Sigma_Y = F \Sigma_T F^\top$, and the least leakage among such descriptions is

$$L_{1d}(D) = \frac{1}{2} \log \left(1 + \frac{\Delta}{\mu_{\max}(\Delta)} \right), \quad (7)$$

where $\mu_{\max}(\Delta) > 0$ is the largest root of $\det(A(\Delta) - \mu \Sigma_{T|S}) = 0$. It is attained with b a corresponding generalized eigenvector and $\sigma^2 = b^\top \Sigma_T Q \Sigma_T b / \Delta - b^\top \Sigma_T b$, at which the distortion constraint is active.

(b) Suppose $\Delta < \lambda_{\max}(W^{1/2} \Sigma_Y W^{1/2})$ and that $\mu_{\max}(\Delta)$ is simple (if it is not, two distinct one-dimensional optima exist, their midpoint is strictly better by Theorem 4, and $L(D) < L_{1d}(D)$). Let Σ_{e0} and Σ_e be the error covariances of the description of (a), and let $\Gamma(\varpi) = \frac{1}{2} \Sigma_e - \varpi \Sigma_{e0} Q \Sigma_{e0}$. Then $L(D) = L_{1d}(D)$ if and only if there is $\varpi \geq 0$ with $\Gamma(\varpi) \succeq 0$ and $\Gamma(\varpi)b = 0$, and then $\varpi = \frac{1}{2} b^\top \Sigma_e b / (b^\top \Sigma_{e0} Q \Sigma_{e0} b)$.

(c) If the reproduction is scalar ($m = 1$), then $L(D) = L_{1d}(D)$ for every $0 < D < D_{\max}$.

Proof. (a) For the description Z , $\text{Cov}(T | Z) = \Sigma_T - \Sigma_T b b^\top \Sigma_T / (b^\top \Sigma_T b + \sigma^2)$, so the distortion of $\mathbb{E}[Y | Z]$ is $D_{\max} - b^\top \Sigma_T Q \Sigma_T b / (b^\top \Sigma_T b + \sigma^2)$, and the constraint reads $\sigma^2 \leq b^\top \Sigma_T Q \Sigma_T b / \Delta - b^\top \Sigma_T b$. The leakage $I(T; Z | S) = \frac{1}{2} \log(1 + b^\top \Sigma_{T|S} b / \sigma^2)$ decreases in σ^2 , so at the largest admissible σ^2 , where the constraint is active, it equals $\frac{1}{2} \log(1 + \Delta b^\top \Sigma_{T|S} b / b^\top A(\Delta) b)$, finite exactly when $b^\top A(\Delta) b > 0$. Minimizing over b is maximizing the Rayleigh quotient $b^\top A(\Delta) b / b^\top \Sigma_{T|S} b$ of a symmetric pencil with $\Sigma_{T|S} \succ 0$, whose maximum is $\mu_{\max}$. Some b has $b^\top A(\Delta) b > 0$ if and only if $\Sigma_T^{1/2} Q \Sigma_T^{1/2} \not\preceq \Delta I$, and the nonzero eigenvalues of that matrix are those of $W^{1/2} \Sigma_Y W^{1/2}$.

(b) The description of (a) has an active constraint and $\Sigma_e^{-1} = \Sigma_{T|S}^{-1} + b b^\top / \sigma^2$, so $\Sigma_{T|S} - \Sigma_e$ is a positive multiple of $\Sigma_{T|S} b b^\top \Sigma_{T|S}$ and $\Sigma_e^{-1} \Sigma_{T|S} b$ is a positive multiple of b . With $\Theta = \Sigma_e^{-1} \Gamma(\varpi) \Sigma_e^{-1}$, the conditions (4) at this point read $\Gamma(\varpi) \succeq 0$ and $\Gamma(\varpi)b = 0$, and Theorem 4(c) makes them necessary and sufficient. Taking $b^\top \Gamma(\varpi) b = 0$ gives ϖ .

(c) By Lemma 1 the optimum is attained by a jointly Gaussian channel $\hat{Y} = a^\top T + N$. If $\text{Cov}(Y, \hat{Y}) = 0$ its distortion is at least $D_{\max} > D$; otherwise $\mathbb{E}[Y | \hat{Y}]$ is a nonzero multiple of $\hat{Y}$, which leaves the leakage unchanged and does not raise the distortion, so the optimum is a one-dimensional description. $\square$

Corollary 9. Let $T = (Y, V)$ be scalar with unit variances and correlation ρ , $|\rho| < 1$, let $W = 1$, and let $S = V + U$ with $\text{Var } U = \tau^2 > 0$; write $s = 1 + \tau^2$. For $0 < D < 1$,

$$L(D) = \frac{1}{2} \log g^*, \quad g^* = \frac{(D + s - \rho^2) + \sqrt{(D + s - \rho^2)^2 - 4Ds(1 - \rho^2)}}{2Ds},$$

the larger root of $P(g) = Dsg^2 - (D + s - \rho^2)g + (1 - \rho^2)$, attained by $\hat{Y} = aY + bV + N$ with $a = (g^* - 1)/g^*$, $b = (g^* - 1)\rho/(g^*(g^*s - 1))$, and the noise variance that makes the distortion equal to D .

Proof. By Theorem 8(c), $L(D) = \frac{1}{2} \log(1 + (1 - D)/\mu_{\max})$ with $\Delta = 1 - D$. Substituting $\mu = (1 - D)/(g - 1)$ into $\det(A(1 - D) - \mu \Sigma_{T|S})$, with $\Sigma_T = \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix}$, $Q = e_1 e_1^\top$, and $J = e_2 e_2^\top / \tau^2$, gives $-(1 - D)(1 - \rho^2)P(g)/(s(g - 1)^2)$, a direct computation confirmed by computer algebra (Appendix C). Since $P(1) = (D - 1)\tau^2 < 0$ and $P(0) = 1 - \rho^2 > 0$, P has one root in $(0, 1)$, which corresponds to a negative μ , and one root $g^* > 1$, which corresponds to the unique positive root $\mu_{\max}$. The read direction is the corresponding generalized eigenvector; normalizing the reproduction gives the stated coefficients. $\square$

For a scalar described variable with vector context, Theorem 8(c) likewise gives $L(D)$ as an explicit function of the largest generalized eigenvalue of a $p \times p$ pencil, $p = 1 + \dim V$.

VI. DIRECTIONS AND MISALIGNMENT

Proposition 10. *With no holder ($J = 0$), the best one-dimensional description reads the top generalized eigenvector of the pencil $(\Sigma_T Q \Sigma_T, \Sigma_T)$ at every budget, and the rate-optimal description reverse water-fills on the eigenvalues of $W^{1/2} \Sigma_Y W^{1/2}$ along fixed directions.*

Proof. With $J = 0$, $\Sigma_{T|S} = \Sigma_T$ and $A(\Delta)b = \mu \Sigma_T b$ reads $\Sigma_T Q \Sigma_T b = (\mu + \Delta) \Sigma_T b$, whose eigenvectors do not depend on Δ . The rate-optimal description is the weighted quadratic rate-distortion solution [15]: in the coordinates $W^{1/2} Y$ the distortion is squared error, and reverse water-filling describes fixed principal components [3]. $\square$

The Gaussian information bottleneck has the same property: its optimal projections are fixed eigenvectors, and its tradeoff parameter sets only their number and scale [4]. The leakage-optimal description behaves differently.

Theorem 11. *Let $\mu_{\max}(\Delta)$ be simple and positive on an open interval $\mathcal{I} \subseteq (0, \lambda_{\max}(W^{1/2} \Sigma_Y W^{1/2}))$, with generalized eigenvector $b(\Delta)$. Then $b(\Delta)$ spans the same line for every $\Delta \in \mathcal{I}$ if and only if, for some $\Delta_0 \in \mathcal{I}$, $b_0 = b(\Delta_0)$ satisfies*

$$Q \Sigma_T b_0 = \lambda b_0 \quad \text{and} \quad J \Sigma_T b_0 = \kappa b_0 \quad (8)$$

for scalars $\lambda > 0$ and $\kappa \geq 0$, that is, if and only if b_0 is a common generalized eigenvector of the pencils $(\Sigma_T Q \Sigma_T, \Sigma_T)$ and $(\Sigma_T J \Sigma_T, \Sigma_T)$. Otherwise $b(\Delta)$ is constant on no open subinterval of $\mathcal{I}$.

Proof. A simple eigenvalue of a symmetric pencil with positive definite second matrix, and its eigenvector, are analytic in the parameter [21]. Write $A(\Delta) = A_0 - \Delta \Sigma_T$.

Suppose b is constant on an open subinterval. Differentiating $(A_0 - \Delta \Sigma_T)b = \mu_{\max}(\Delta) \Sigma_{T|S} b$ gives $\Sigma_T b = c \Sigma_{T|S} b$ for a scalar c , and $c = b^\top \Sigma_T b / b^\top \Sigma_{T|S} b \geq 1$ because $\Sigma_{T|S} \preceq \Sigma_T$. Then $A_0 b = (\Delta + \mu_{\max}/c) \Sigma_T b$, that is, $Q \Sigma_T b = \lambda b$ with $\lambda = \Delta + \mu_{\max}/c > 0$; and multiplying $\Sigma_T b = c \Sigma_{T|S} b$ by $\Sigma_{T|S}^{-1} = \Sigma_T^{-1} + J$ gives $J \Sigma_T b = (c - 1)b$, so $\kappa = c - 1 \geq 0$. This proves the last sentence and the forward direction.

Conversely, if b_0 satisfies (8), then $\Sigma_{T|S}^{-1} \Sigma_T b_0 = (1 + \kappa) b_0$, so $\Sigma_T b_0 = (1 + \kappa) \Sigma_{T|S} b_0$ and $A(\Delta) b_0 = (\lambda - \Delta) \Sigma_T b_0 = (\lambda - \Delta)(1 + \kappa) \Sigma_{T|S} b_0$ for every Δ : b_0 is a generalized eigenvector with eigenvalue $(\lambda - \Delta)(1 + \kappa)$ throughout. Let $\mathcal{E} = \{\Delta \in \mathcal{I} : (\lambda - \Delta)(1 + \kappa) = \mu_{\max}(\Delta)\}$. It contains Δ_0 and is closed in $\mathcal{I}$ by continuity. It is open, because near a point of $\mathcal{E}$ the simple eigenvalue $\mu_{\max}$ is the only eigenvalue of the pencil close to it, and $(\lambda - \Delta)(1 + \kappa)$ is an eigenvalue close to it. Since $\mathcal{I}$ is connected, $\mathcal{E} = \mathcal{I}$, and by simplicity $b(\Delta)$ spans the line of b_0 throughout. $\square$

Corollary 12. *Suppose the holder observes only the context: in coordinates $T = (Y, V)$, $F = [I \ 0]$ and $H = [0 \ H_V]$. Then (8) holds if and only if $b_0 = F^\top \beta$ with $W \Sigma_Y \beta = \lambda \beta$ and $\text{Cov}(S, \beta^\top Y) = 0$.*

Proof. $Q \Sigma_T b_0 = \lambda b_0$ with $\lambda > 0$ puts b_0 in the range of Q , which is the range of $F^\top$, so $b_0 = F^\top \beta$ and $F^\top W \Sigma_Y \beta = \lambda F^\top \beta$. The vector $J \Sigma_T b_0$ lies in the range of $H^\top$, which meets the range of $F^\top$ only at zero, so $J \Sigma_T b_0 = \kappa b_0$ forces $J \Sigma_T b_0 = 0$; with $w = \Sigma_T b_0$, $w^\top J w = 0$ gives $\Sigma_U^{-1/2} H w = 0$, that is, $\Sigma_U^{-1} H \Sigma_T F^\top \beta = 0$, which is $\text{Cov}(S, \beta^\top Y) = 0$. The converse is direct. $\square$

The same condition decides whether the two optimal descriptions coincide. Let b_R be the top generalized eigenvector of $(\Sigma_T Q \Sigma_T, \Sigma_T)$, the read direction of the best one-dimensional rate description.

Theorem 13. *Let $\lambda_1 \geq \lambda_2$ be the two largest eigenvalues of $W^{1/2} \Sigma_Y W^{1/2}$ and suppose $\Delta \leq \lambda_1 - \lambda_2$, the regime in which the rate-optimal description is one-dimensional, $b_R^\top T + N$. If $J \Sigma_T b_R$ is not parallel to b_R , the rate-optimal description is not leakage-optimal and the leakage-optimal description is not rate-optimal, so each pays a strictly positive excess in the other cost. If $J \Sigma_T b_R$ is parallel to b_R , then b_R is a generalized eigenvector of the leakage pencil at every Δ , and inside the one-dimensional regime of Theorem 8 the two descriptions coincide exactly when it is the top one.*

Proof. In the stated regime reverse water-filling describes only the principal component, so the rate-optimal description is the one-dimensional description reading b_R , where $Q \Sigma_T b_R = \lambda_1 b_R$. If it were leakage-optimal, then $L(D) = L_{1d}(D)$ and b_R would maximize the Rayleigh quotient of Theorem 8(a), hence be a generalized eigenvector of $(A(\Delta), \Sigma_{T|S})$, that is, $(\lambda_1 - \Delta) \Sigma_T b_R \propto \Sigma_{T|S} b_R$; multiplying by $\Sigma_{T|S}^{-1} = \Sigma_T^{-1} + J$ gives $J \Sigma_T b_R \propto b_R$. So when $J \Sigma_T b_R$ is not parallel to b_R the rate-optimal description is not leakage-optimal, and since both minimizers of

Theorem 4(a) are unique, the leakage-optimal description is not rate-optimal either. The same computation read backward shows that $J\Sigma_T b_R \propto b_R$ makes b_R a generalized eigenvector at every Δ , and inside the one-dimensional regime the leakage-optimal description reads the top one. $\square$

In the scalar case of Corollary 9, $b_R = e_1$ and $J\Sigma_T e_1 = (\rho/\tau^2)e_2$, which is parallel to e_1 only when $\rho = 0$. So the two descriptions differ exactly when the context is correlated with the described variable. The rate-optimal channel is the classical reverse channel, with leakage $\frac{1}{2} \log[(1-D)(1-\rho^2/s) + D]/D$, strictly above $L(D)$, and the leakage-optimal channel of Corollary 9 has rate strictly above $\frac{1}{2} \log(1/D)$. The rate excess is unbounded toward the clean-context boundary $\tau^2 \rightarrow 0$ at $D = 1 - \rho^2$, where the leakage-optimal channel pins the reproduction to the context.

Corollary 14. *There are sources for which the leakage-optimal one-dimensional read takes different directions at two budgets. For such a source no description problem for a single party with a fixed weighted quadratic distortion has the leakage-optimal descriptions as its optimal descriptions at both budgets.*

Proof. Take the scalar source of Corollary 9 with $\rho \neq 0$. By Theorem 8(c) the leakage-optimal description is the one-dimensional description at every D , and its read direction is proportional to $(g^*s - 1, \rho)$, the ratio of the coefficients in Corollary 9. Since ℓ is strictly decreasing (Lemma 15), g^* is strictly decreasing in D , so the read line differs at any two budgets. By Proposition 10, applied to any fixed weight on T , the best one-dimensional description for a single party reads the same line at both budgets. The coupled instance of Section VIII, whose leakage-optimal descriptions reduce the error along 47.48° and 41.15° at two budgets, illustrates the same effect in the space of the described variable. $\square$

VII. INDEPENDENT COMPONENTS

Suppose $Y = (Y_1, \dots, Y_m)$, W is diagonal with entries w_j , and each Y_j has its own context V_j and holder observation $S_j = V_j + U_j$, the triples (Y_j, V_j, S_j) being independent, with $\text{Var } Y_j = \sigma_j^2$, $\text{Var } V_j = 1$, $\mathbb{E}[Y_j V_j] = \rho_j \sigma_j$, $|\rho_j| < 1$, and $\text{Var } U_j = \tau_j^2 > 0$. Absorb w_j into σ_j^2 . Write ℓ_j for the scalar function $\delta \mapsto \frac{1}{2} \log g^*(\delta)$ of Corollary 9 at (ρ_j, τ_j^2) , extended by zero for $\delta \geq 1$, and $L_j(D_j) = \ell_j(D_j/\sigma_j^2)$.

Lemma 15. *For $0 < \delta < 1$, with $g = g^*(\delta)$, $s = 1 + \tau^2$, $k = gs - 1$, and $\Delta_\delta = \sqrt{(\delta + s - \rho^2)^2 - 4\delta s(1 - \rho^2)} > 0$,*

$$\ell'(\delta) = -\frac{k}{2 \ln 2 \Delta_\delta}, \quad \ell''(\delta) = \frac{(1 - \rho^2)k^3 + \rho^2 \tau^2 g^2 (\tau^4 g + \tau^2 (2g + 1) + g + 1)}{2 \ln 2 g \Delta_\delta^3} > 0,$$

so ℓ is strictly convex and decreasing, $\ell' \rightarrow -\infty$ as $\delta \rightarrow 0$, and $\ell' \rightarrow -\tau^2/(2 \ln 2(\rho^2 + \tau^2))$ as $\delta \rightarrow 1$.

Proof. The larger root is simple, with $\partial_g P = \Delta_\delta$ there, so g^* is smooth; differentiating $P(g^*(\delta)) = 0$ with $\partial_\delta P = gk$ gives $g' = -gk/\Delta_\delta$ and the first formula. Differentiating once more and eliminating δ through $\delta gk = (s - \rho^2)g - (1 - \rho^2)$ gives the second, an identity of polynomials confirmed by computer algebra (Appendix C); its numerator is positive. The limits follow from $g^* \rightarrow \infty$, $\Delta_\delta \rightarrow s - \rho^2$ as $\delta \rightarrow 0$ and $g^* \rightarrow 1$, $k \rightarrow \tau^2$, $\Delta_\delta \rightarrow \rho^2 + \tau^2$ as $\delta \rightarrow 1$. $\square$

Define the activation level $\kappa_j = -L'_j(\sigma_j^2) = \tau_j^2/(2 \ln 2 \sigma_j^2(\rho_j^2 + \tau_j^2))$; for the rate it would be $1/(2 \ln 2 \sigma_j^2)$.

These components are two-dimensional blocks of Theorem 5, each solved by Corollary 9.

Theorem 16. (a) $\mathcal{R}(D)$ is the union, over $D_j \geq 0$ with $\sum_j D_j \leq D$, of the Minkowski sums of the scalar regions of the components, attained by product channels. (b) $L(D) = \min \sum_j L_j(D_j)$ over such allocations, with a unique minimizer $D_j = D_j(\theta^*)$, where $D_j(\theta) = \sigma_j^2$ for $\theta \leq \kappa_j$ and otherwise solves $-L'_j(D_j) = \theta$, and θ^* is the unique level with $\sum_j D_j(\theta^*) = D$. (c) As D decreases, the leakage-optimal description begins to describe component j in decreasing order of $\sigma_j^2(1 + \rho_j^2/\tau_j^2)$, while the rate-optimal description proceeds in decreasing order of σ_j^2 .

Proof. (a) By Theorem 3 it suffices to identify the single-letter union. For an admissible channel with $D_j = \mathbb{E}(Y_j - \hat{Y}_j)^2$, each induced channel $p(\hat{y}_j | t_j)$ satisfies $\hat{Y}_j - T_j - S_j$, and since the pairs (T_j, S_j) are independent the chain rule gives $I(T; \hat{\mathbf{Y}}) \geq \sum_j I(T_j; \hat{Y}_j)$ and $I(T; \hat{\mathbf{Y}} | S) = \sum_j [I(T_j; \hat{\mathbf{Y}}, S, T^{j-1}) - I(T_j; S_j)] \geq \sum_j I(T_j; \hat{Y}_j | S_j)$. A product of component channels attains the sum with equality. (b) Each L_j is convex and nonincreasing, strictly convex on $(0, \sigma_j^2)$ by Lemma 15, with left derivative $-\kappa_j$ at σ_j^2 . An allocation is optimal if and only if some

$\theta \geq 0$ lies in $-\partial L_j(D_j)$ for every j with $\theta(\sum_j D_j - D) = 0$. Since $D < \sum_j \sigma_j^2$, $\theta > 0$; no D_j is 0 because $-L'_j \rightarrow \infty$ there; and $\sum_j D_j(\theta)$ decreases continuously and strictly from $\sum_j \sigma_j^2$ at $\min_j \kappa_j$ to 0, so θ^* is unique. (c) $D_j(\theta) < \sigma_j^2$ exactly when $\theta > \kappa_j$, and θ^* increases as D decreases. $\square$

Corollary 17. *If $m = 2$, $\sigma_1^2 > \sigma_2^2$, and $\sigma_1^2(1 + \rho_1^2/\tau_1^2) < \sigma_2^2(1 + \rho_2^2/\tau_2^2)$, then for every $D \in [\max\{2\sigma_2^2, \sigma_1^2 + D_2(\kappa_1)\}, \sigma_1^2 + \sigma_2^2]$ the rate-optimal description describes component 1 alone and the leakage-optimal description describes component 2 alone.*

Proof. Reverse water-filling describes component 1 alone exactly for $2\sigma_2^2 \leq D < \sigma_1^2 + \sigma_2^2$; Theorem 16(b) describes component 2 alone exactly when $\kappa_2 < \theta^* \leq \kappa_1$, that is, for $\sigma_1^2 + D_2(\kappa_1) \leq D < \sigma_1^2 + \sigma_2^2$. $\square$

VIII. INSTANCES

Scalar. At $(\rho^2, \tau^2, D) = (0.75, 0.5, 0.3)$ the rate-optimal channel pays 0.0349 bits of excess leakage and the leakage-optimal channel 0.0400 bits of excess rate; over the box $\rho^2 \in [0.05, 0.95]$, $\tau^2 \in [\frac{1}{4}, 4]$, $D \in [0.05, 0.95]$ the rate excess of the leakage-optimal channel stays below 0.114 bits and the leakage excess of the rate-optimal channel below 0.077 bits, and at $(\rho^2, \tau^2, D) = (0.5, 10^{-3}, 0.5)$ the rate excess reaches 1.537 bits. Two sources with the same law of (Y, S) up to scaling of S , $(\rho^2, \tau^2) = (\frac{3}{5}, \frac{1}{5})$ and $(\frac{3}{4}, \frac{1}{2})$, have $L = 1.1880$ and 1.2105 bits at $D = 0.1$: the leakage is not determined by the law of (Y, S) , because the encoder observes the context.

Coupled. Take $\Sigma_Y = \text{diag}(1, 0.4)$, $W = I$, a scalar context with $\text{Var } V = 1$ and $\text{Cov}(Y, V) = 0.6u$ for the unit vector u at 50° , and $S = V + U$ with $\text{Var } U = 0.05$. The certificate of Theorem 8(b) holds for $D \geq 1.08530$, out of $D_{\max} = 1.4$. The rate-optimal description reduces the error along the first axis for every $D \geq 0.8$. The leakage-optimal description reduces it along 47.48° , 46.76° , 45.07° , and 41.15° at $D = 1.38, 1.3, 1.2$, and 1.1 . Four sources with the same u and the same law of (Y, S) up to scaling, $(|\text{Cov}(Y, V)|^2, \tau^2) = (0.33, 0.1)$, $(0.36, 0.2)$, $(0.39, 0.3)$, $(0.42, 0.4)$, read along 42.83° , 37.61° , 33.75° , and 30.84° at $D = 1.3$, each inside its certificate regime: the direction, too, is not determined by the law of (Y, S) .

Independent components. With $(\sigma_1^2, \rho_1^2, \tau_1^2) = (1, 0, 1)$ and $(\sigma_2^2, \rho_2^2, \tau_2^2) = (0.6, 0.9, 0.05)$, $\kappa_1 = 0.7213$, $\kappa_2 = 0.0633$, and the two descriptions describe disjoint components for every $D \in [1.2002, 1.6)$. At $D = 1.4$ the rate-optimal description pays $R = L = 0.1610$ bits on component 1 and the leakage-optimal description pays $L = 0.0197$ and $R = 0.3272$ bits on component 2.

IX. DISCUSSION

A description held by two parties has a cost for each, and the encoder's observations decide how the costs trade off. For Gaussian sources without decoder side information the region is now explicit in structure: a convex program in the error covariance, a closed form in a one-dimensional regime, and directions that turn with the budget exactly when the two parties' geometries are not aligned. Several problems remain open. With side information at the decoder the weighted problem of Ekrem and Ulukus was left open there, and a solution is reported by Xu and Chen [5]; it is harder than the present case because the Gaussian exhaustion lemma no longer handles both costs at once once the decoder bins against its own observation. Explicit solutions are now available in two complementary regimes, commuting geometry at every budget and arbitrary geometry in the one-dimensional regime; a closed form for non-commuting geometry below the one-dimensional regime, and for the regime boundary, is not known. Stationary sources and finite blocklengths are not treated.

APPENDIX A THE CODING THEOREM

A. Finite alphabets

Let (X, S) be a discrete memoryless pair with finite alphabets and let d be a distortion with values in $[0, d_{\max}]$; codes, rate, distortion, and leakage $H(M | S^n)/n$ are as in Section II.

Theorem 18 ([1, Theorem 3], without decoder side information). *The closure of the achievable set at distortion D is $\bigcup_{p(\hat{x}|x): \mathbb{E}d(X, \hat{X}) \leq D} \{(R, L) : R \geq I(X; \hat{X}), L \geq I(X; \hat{X} | S)\}$, and the union is convex and closed.*

Proof outline. Villard and Piantanida prove the region in the equivocation form $\Delta \leq H(X | V) + 0 - I(X; S | U)$, $U - V - X - S$, with no decoder side information; taking $U = V$ is optimal, because $I(X; S | U) = I(V; S |$

$U) + I(X; S | V) \geq I(X; \hat{S} | V)$, and $\Delta = H(X | S) - L$ for a deterministic encoder, which gives the displayed form with V in place of $\hat{X}$, and a reproduction that is a function of V only lowers both coordinates. For the Gaussian extension we use the following finite-blocklength form of the achievability, obtained by covering at rate $I(X; \hat{X}) + 2\epsilon$, a random binning of the codeword index into $2^{n(I(X; \hat{X}|S) + 3\epsilon)}$ bins used only as a proof device, Fano's inequality for a hypothetical decoder holding the bin index and S^n , and selection of one deterministic code: for every channel with finite reproduction alphabet and every $\epsilon > 0$ there are $\epsilon_n \rightarrow 0$ and, for all large n , a deterministic code with $|\mathcal{M}_n| \leq 2^{n(I(X; \hat{X}) + 2\epsilon)}$, distortion at most $\mathbb{E}d(X, \hat{X}) + \epsilon + 2\epsilon_n d_{\max}$, and $H(M | S^n) \leq n(I(X; \hat{X} | S) + 3\epsilon) + 1 + 2\epsilon_n n(I(X; \hat{X}) + 2\epsilon)$. Convexity of both coordinates in the channel, for the conditional one by the decomposition $I(X; \hat{X} | S) = \sum_s p(s) I_{p(x|s)}(q)$, makes the union convex without time sharing. $\square$

B. The Gaussian source

Proof of Theorem 3. Converse. For any code, $\log |\mathcal{M}_n| \geq I(T^n; \hat{\mathbf{Y}}^n) \geq \sum_i I(T_i; \hat{\mathbf{Y}}_i)$ and $H(M | S^n) \geq I(T^n; \hat{\mathbf{Y}}^n | S^n) = \sum_i [I(T_i; \hat{\mathbf{Y}}^n, T^{i-1}, S^n) - I(T_i; S_i)] \geq \sum_i I(T_i; \hat{\mathbf{Y}}_i | S_i)$, by the chain rule and the independence of the pairs (T_i, S_i) ; every term is finite, being at most $I(T_i; S_i) + \log |\mathcal{M}_n|$ with $I(T_i; S_i) = \frac{1}{2} \log \det(H\Sigma_T H^\top + \Sigma_U) / \det \Sigma_U$. The mixture of the per-letter channels satisfies the Markov chain, has distortion D_n , and by convexity of both coordinates in the channel (for general alphabets the conditional coordinate is an integral over s of relative entropies affine in the channel [25], [24]) lies below (R_n, L_n) , so (R_n, L_n) is in the union at D_n .

Closedness. Let $(R_j, L_j) \rightarrow (R, L)$ lie in the union at levels D_j with $\limsup D_j \leq D$, with $\Sigma_j \in \mathcal{P}(D_j)$ below them by Proposition 2. The Σ_j lie in the compact set $0 \preceq \Sigma \preceq \Sigma_T$ and $\det \Sigma_j \geq \det \Sigma_T 2^{-2R_j}$ is bounded away from zero, so a subsequence converges to some $\Sigma^* \succ 0$ in $\mathcal{P}(D)$; r and ℓ are continuous on positive definite matrices, so (R, L) lies above $(r(\Sigma^*), \ell(\Sigma^*))$.

Achievability. Fix an admissible channel with $I(T; \hat{\mathbf{Y}}) < \infty$; $\mathbb{E}\|\hat{\mathbf{Y}}\|^2 < \infty$ because its distortion is finite and $W \succ 0$. Let q_k be the saturating dyadic quantizer of level k applied coordinatewise: the bounded cells $[j2^{-k}, (j+1)2^{-k})$, $-4^k \leq j < 4^k$, map to their left endpoints and the two tail cells to $\pm 2^k$; the partitions are nested and generate the Borel σ -field.

Step A. The σ -fields of $S_\eta := q_\eta(S)$ increase to $\sigma(S)$, so $I(\hat{\mathbf{Y}}; S_\eta) \uparrow I(\hat{\mathbf{Y}}; S)$ [22, Ch. 2]; fix η with $I(\hat{\mathbf{Y}}; S_\eta) \geq I(\hat{\mathbf{Y}}; S) - \epsilon$. Since $\hat{\mathbf{Y}} - T - S_\eta$, $I(T; \hat{\mathbf{Y}} | S_\eta) = I(T; \hat{\mathbf{Y}}) - I(\hat{\mathbf{Y}}; S_\eta) \leq I(T; \hat{\mathbf{Y}} | S) + \epsilon$.

Step B. With $T_k = q_k(T)$ and $\hat{\mathbf{Y}}_k = q_k(\hat{\mathbf{Y}})$, the surrogate source (T_k, S_η) is discrete memoryless, the surrogate channel is $p(\hat{\mathbf{y}} | t_k) = \Pr(\hat{\mathbf{Y}}_k = \hat{\mathbf{y}} | T_k = t_k)$, and the surrogate distortion $d_k(t_k, \hat{\mathbf{y}}) = \mathbb{E}[(Y - \hat{\mathbf{y}})^\top W (Y - \hat{\mathbf{y}}) | T_k = t_k]$ is bounded for each k , every cell having positive probability since $\Sigma_T \succ 0$. The finite-blocklength achievability of Theorem 18 gives a code of rate at most $I(T_k; \hat{\mathbf{Y}}_k) + 2\epsilon \leq I(T; \hat{\mathbf{Y}}) + 2\epsilon$. Its true distortion equals its surrogate distortion, because given T_k^n the law of Y_i depends on $T_{k,i}$ alone. The surrogate distortion of the channel converges to $\mathbb{E}[(Y - \hat{\mathbf{Y}})^\top W (Y - \hat{\mathbf{Y}})] \leq D$, because $\mathbb{E}[Y | T_k] \rightarrow Y$ in L^2 by martingale convergence and $\hat{\mathbf{Y}}_k \rightarrow \hat{\mathbf{Y}}$ in L^2 . The surrogate law of $(\hat{\mathbf{Y}}_k, S_\eta)$ converges weakly to the true law of $(\hat{\mathbf{Y}}, S_\eta)$, by the bounded-convergence and martingale argument applied to test functions bounded on $\mathbb{R}^m \times q_\eta(\mathbb{R}^r)$ and continuous in the first argument, and the Markov chain $\hat{\mathbf{Y}} - T - S$. Mutual information is lower semicontinuous under weak convergence [23], so $I_k(T_k; \hat{\mathbf{Y}}_k | S_\eta) = I(T_k; \hat{\mathbf{Y}}_k) - I_k(\hat{\mathbf{Y}}_k; S_\eta) \leq I(T; \hat{\mathbf{Y}} | S) + 2\epsilon$ for large k , and the code has $H(M | S_\eta^n) \leq n(I(T; \hat{\mathbf{Y}} | S) + 5\epsilon + o(1))$.

Step C. S_η^n is a function of S^n , so $H(M | S^n) \leq H(M | S_\eta^n)$. A diagonal sequence $\epsilon \downarrow 0$ achieves the corner $(I(T; \hat{\mathbf{Y}}), I(T; \hat{\mathbf{Y}} | S))$ at distortion D , and with the converse, closedness, and Proposition 2 the region is as stated. $\square$

APPENDIX B

THE GAUSSIAN EXHAUSTION LEMMA

Proof of Lemma 1. The divergence bound. Conditional mutual information is the expected divergence of the conditional law of T given $(\hat{\mathbf{Y}}, S)$ from the conditional law given S [24], [22]. Fix a realization, let P be the conditional law with mean μ and covariance C , $P_G = \mathcal{N}(\mu, C)$, and $Q_S = \mathcal{N}(\mu_S, \Sigma_{T|S})$. If P has no density the divergence is infinite. Otherwise $\log(dP_G/dQ_S)$ is quadratic in t and P, P_G share two moments, so $D(P||Q_S) = D(P||P_G) + D(P_G||Q_S) \geq D(P_G||Q_S)$. Averaging $D(P_G||Q_S)$, the trace term plus the mean term, $\mathbb{E}[\text{tr}(\Sigma_{T|S}^{-1}C) + (\mu - \mu_S)^\top \Sigma_{T|S}^{-1}(\mu - \mu_S)]$, averages to p by the law of total covariance, and by concavity of $\ln \det$ and Jensen, $\mathbb{E}[-\ln \det C] \geq -\ln \det \mathbb{E}C \geq -\ln \det \Sigma_e$, since the conditional mean has error covariance no larger

than any linear estimator. Hence $I(T; \hat{\mathbf{Y}} | S) \geq \frac{1}{2} \log(\det \Sigma_{T|S} / \det \Sigma_e)$, and the same argument without S gives $I(T; \hat{\mathbf{Y}}) \geq \frac{1}{2} \log(\det \Sigma_T / \det \Sigma_{e0})$.

The determinant identity. With $\hat{\mathbf{Y}} = AT + N'$, $\text{Cov}(N', T) = 0$, and $\text{Cov}(N', S) = 0$, the change of variables $(T, S, \hat{\mathbf{Y}}) \mapsto (T, S, N')$ is block triangular with unit diagonal, so $\det \text{Cov}(T, S, \hat{\mathbf{Y}}) = \det \Sigma_S \det \Sigma_{T|S} \det \Sigma_N$; the Schur factorization in the other order gives $\det \text{Cov}(\hat{\mathbf{Y}}, S) \det \Sigma_e$, and $\text{Cov}(\hat{\mathbf{Y}}) - \text{Cov}(\hat{\mathbf{Y}}, S) \Sigma_S^{-1} \text{Cov}(S, \hat{\mathbf{Y}}) = A \Sigma_{T|S} A^\top + \Sigma_N \succ 0$. So $\det \Sigma_e = \det \Sigma_{T|S} \det \Sigma_N / \det(A \Sigma_{T|S} A^\top + \Sigma_N)$, finite and positive.

Equality. For the jointly Gaussian channel $AT + N$ the conditional law of T given $(\hat{\mathbf{Y}}, S)$ is Gaussian with the linear estimate as mean and Σ_e as covariance, so every inequality is an equality; its second moments match the given channel's. $\square$

APPENDIX C NUMERICAL CHECKS

The identities and instances were checked independently of the derivations by scripts run on a remote workstation. `verify_general.py` checks, for random sources with general weights and observation maps, that the pencil value of Theorem 8 equals the brute-force optimum over Gaussian descriptions whenever that optimum is one-dimensional, that both costs are convex in the error covariance (16000 midpoint tests), that the weighted minimizers are unique, the two directions of Theorem 11, and the criterion of Theorem 13. `verify_commuting.py` checks the closed form of Theorem 6 for the minimum leakage and for weighted frontier points against brute force, the diagonality of the optimum in the common eigenbasis, the activation rule, and the block-diagonal optimum of Theorem 5. `verify_coupled.py`, `verify_parallel.py`, and `verify_pencil.py` check the scalar identity of Corollary 9 symbolically, the determinant program against brute force, the certificate, Lemma 15, the allocation, and the numbers of Section VIII. No proof depends on them; the instances illustrate theorems proved in the text.

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